

## Java VS Javascript

|                                                                                     |                                                                                                              |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Java is a programming language, in which the code will get executed sequentially.   | In Javascript the program is event driven, the execution of code is dependent on user action                 |
| Java is statically type, means the data type of variable is decided at compile time | In javascript type of variables is decided at run time, based on the value you have assigned to the variable |
| Java is Object oriented.                                                            | Javascript is object based.                                                                                  |

To declare a variable in javascript, you can use keyword let, var and const

| let                                             | var                                         | const                                                                            |
|-------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------|
| duplicate declaration is not allowed            | duplicate declaration is allowed            | duplicate declaration is not allowed                                             |
| let is a blocked scope variable                 | var is global or function scope variable    | const is blocked scope variable                                                  |
| hoisting of variable declaration is not allowed | hoisting of variable declaration is allowed | hoisting of variable declaration is not allowed                                  |
| let v;                                          | var v;                                      | const c=34;<br>Initialization of variable is needed, at the time of declaration. |

In client side javascript

1. Client side data validation is possible
2. Give dynamic look and feel to the page
3. AJAX(Asynchronous Javascript and XML) requests can be send by javascript to webservice, by using XmlHttpRequest(XHR) object

The Javascript code can be executed with the help of HTML, and is called as Client side javascript code.

Or it can be executed with the help of Nodejs, which is server side code.

To display popup boxes in javascript, there are 3 types of boxes, and all these functions are of window objects

| Popup boxes |                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| alert       | A box with only one ok button, useful to show messages to user                                                                                          |
| confirm     | A box with 2 buttons ok and cancel, if user clicks on ok button then it returns true, otherwise, if user clicks on cancel button, then it returns false |
| prompt      | A box with a text box, useful to accept one input from user                                                                                             |

To display ouput to user in browser window, document.write(“Hello world!!”)

To display messages for testing purpose the use console.log(“Test messages”);

## Ways to write functions

|                                                                                                                                                       |                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>const f3=(x,y=3,...s)=&gt;{//rest parameter     document.write("in f3",x,y,"&lt;br&gt;") } f1(1,2,3,4) f2(1,2) f3(1,2,3,4)</pre>                 | <p>function using =&gt; operator, you may assign default values to parameters<br/>It doesnot have arguments array</p>                                                         |
| <pre>function f1(x,...arr){     document.write("in f1",x,"&lt;br&gt;");     document.write("test:" +arr.length,arguments[0]); }</pre>                 | <p>... rest parameters are useful for variable number of arguments</p>                                                                                                        |
| <pre>const f2=function(x,y){     document.write("in f2",x,y,"&lt;br&gt;"); } //var f2=45; const f4=(x,y=45,z=4)=&gt;{     console.log(x,y,z); }</pre> | <p>better to use const keyword while declaring function, so that definition will not get overwritten</p>                                                                      |
| <pre>function f1(){     function f2(){     } }</pre>                                                                                                  | <p>Closure function→the function which has access to its parent scope, is called as closure function</p>                                                                      |
| <pre>(function(x,y){     document.write("in function f1",x,y) })(12,13)</pre>                                                                         | <p>Self calling function, If you want to execute the function definition only once, then use self calling function<br/>Usually these functions are used in JQuery plugins</p> |

## functions in javascript for typecasting

|            |                                    |
|------------|------------------------------------|
| parseInt   | to convert the string into integer |
| parseFloat | To convert the string into float   |

The tags which has innerHTML property-→ div, span, p, a,h1,h2....

The tags which are not used to accept data from user via form, use innerHTML property  
but the tags which are used in the form to accept data from user has value property

## functions on numeric data

|                |                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------|
| isNaN(data)    | if the data contains atleast one character, then it returns true, otherwise it returns false  |
| isFinite(data) | if the data contains all digits between 0-9, then it returns true, otherwise it returns false |

### Validate data in javascript

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| validatedropdown | <p>In dropdown box it stores property by name selectedIndex, it stores the index position of the selected option</p> <p>It also stores a array by name options, hence to find the value of selected dropdown box in javascript</p> <pre>//get reference of dropdown box centers var c=document.getElementById(id); //retrieve the value of index of the selected option var pos=c.selectedIndex; //alert(c.selectedIndex) //retrieve the value of selected option var val=c.options[pos].value;</pre>                                                   |
| validateradio    | <p>to find which radio button is checked we use checked property, In radio button we can select only one option, so as soon as we get the checked radio button we may break the loop</p> <pre>const validateradio=(nm,errid)=&gt;{     var arr=document.getElementsByName("gender");     for(var i=0;i&lt;arr.length;i++){         if(arr[i].checked){             document.getElementById(errid).innerHTML="";             return true;         }     }     document.getElementById(errid).innerHTML="pls select one choice"     return false; }</pre> |
| validatecheckbox | <p>to find which check is checked, we use checked property, checkboxes are used to select multiple options, we use the checked property to find which checkboxes are checked</p> <pre>const validatecheckbox=(nm,errid)=&gt;{     var subarr=document.getElementsByName(nm);     var cnt=0;     var str="";     for(var i=0;i&lt;subarr.length;i++){         if(subarr[i].checked){             cnt++;             str=str+subarr[i].value+"";         }         /*if(cnt&gt;=2){             break;         }*/     }     if(cnt&gt;=2){</pre>         |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <pre>//clear the message document.getElementById(errid).innerHTML=""; alert(str); return true; } //display error message document.getElementById(errid).innerHTML="pls select minimum 2 subjects" return false; }</pre>                                                                                                                                                                                                                                                                                                  |
| validatefile | <p>to validate file size, we may use size property</p> <pre>function validateSize(){     alert("in validatesize")     var f=document.getElementById("files");     //f.files[0].size it will give size in bytes     alert(f.files[0].size/1024+"KB");     if(f.files[0].size/1024 &lt; 10 ){         alert("in if")         document.getElementById("errfile").innerHTML="";         //return true;     }     document.getElementById("errfile").innerHTML="file size exceeds the limit 10KB"     //return false; }</pre> |

## operators

|                      |                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arithmetic operators | <p>+, -, *, / , % , **(exponential)</p> <p>3**3=27 (3 raise to 3)</p>                                                                                                                                   |
| relational operator  | <p>&gt;, &lt;, &lt;=, &gt;=, == , ===, !=, !==</p> <p>=== → strict checking</p> <p>10===10 false //it check value as well as data type</p> <p>10===10 true</p> <p>10=="10" true</p> <p>10===10 true</p> |
| logical operators    | && ,    , !                                                                                                                                                                                             |

## Functions

|                 |                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------|
| a.toFixed(num)  | <p>to display the number upto 2 decimal places</p> <p>a=3.3333333</p> <p>a.toFixed(2)= 3.33</p> |
| Math.round(num) | <p>it will round the number</p> <p>Math.round(3.456778)= 3</p> <p>Math.round(3.6789)=4</p>      |

## String functions

|                                |                                                                                          |                                                                                                                                     |
|--------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| s.trim()                       | to remove leading and trailing spaces                                                    | s=" aaaaa "<br>s.trim()<br>aaaaa                                                                                                    |
| s.length                       | to find length of the string                                                             | s.length                                                                                                                            |
| s.charAt(i)                    | to find the character at specific index position                                         | s="testing"<br>s.charAt(4)=i                                                                                                        |
| s.concat(str1,str2,.....)      | to concat multiple strings                                                               | s="taste"<br>s.concat("is", "good")<br>taste is good                                                                                |
| s.indexOf(str)                 | to find the position of first occurrence of the given str, indexing starts with 0        | s="Hello World!!"<br>s.indexOf("Hello") //0<br>s.indexOf("World") //6                                                               |
| s.split(delimiter)             | to break the string into array of strings at delimiter position                          | s="rain in spain, is pain"<br>s.split(" ")<br>["rain", "in", "spain", "is", "pain"]<br>s.split(",")<br>["rain in spain", "is pain"] |
| s.toUpperCase()                | convert the string into uppercase                                                        | s="abcd"<br>s.toUpperCase()<br>ABCD                                                                                                 |
| s.toLowerCase()                | convert the string into lowercase                                                        | s="ABcd"<br>s.toLowerCase()<br>abcd                                                                                                 |
| s.slice(startIndex[,endIndex]) | it retrieves the portion of the string from startIndex to endIndex, endIndex is excluded | s="Hello world!!"<br>s.slice(0,1) //H<br><br>s.slice(4) //o world!!                                                                 |
| s.includes(substr)             | it returns true, if substring exists, false otherwise                                    | s="Hello world!!"<br>s.includes("ell") //true<br><br>s.includes("xxx") //false                                                      |

## using regular expression

|                                       |                                                                                |                                                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| s.match(regular expression)           | it checks the specified value in the string, and return <b>all occurrences</b> | s="rain in SPAIN is pain"<br>pattern=/a.*?n/<br>s.match(pattern) //2<br><br>pattern=/a.*?n/i -> values enclosed in // is treated as regular expression<br>i is flag for ignore index<br>g is a flag for global<br>s.match(pattern) //2 |
| s.replace(regexexpression, newstring) | It will search the given pattern and replaces it with the newstring            | s="To be or not to be"<br>s.replace(/be/, "exists") -> replace first occurrence                                                                                                                                                        |

|                           |                                                                                            |                                                                                                                                                               |
|---------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                                                                            | s.replace(/be/g,"exists")-><br>flag g, will replace all occurrences                                                                                           |
| s.search(regexexpression) | it checks the specified pattern in the string, and return the position of first occurrence | var test="testing: 1, 2, 3"<br>var pattern= /\d+/g<br>s.search(pattern) //9<br>s.match(pattern) //["1","2","3"]<br>s.replace(pattern,"#") // testing: #, #, # |